

The Caucasian Hunter-Gatherer Linguistic World:

**A Unified Hypothesis on the Origins of
Hurro-Urartian,**

Kartvelian, and Tyrsenian Language Families

A Multidisciplinary Study in Historical Linguistics and Archaeogenetics

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Abstract

This paper proposes a unified framework for understanding three ancient language families — Hurro-Urartian, Kartvelian, and Tyrsenian (Etruscan/Lemnian/Raetic) — as descendants of a common Caucasian Hunter-Gatherer (CHG) linguistic ancestor dating to the Mesolithic period. Drawing on comparative personal pronoun systems, grammatical architecture, phonological inventories, and recent archaeogenetic evidence, we argue that

these families share a set of structural features — pure suffixing morphology, ergative-absolutive alignment, animacy-based nominal distinctions, Suffixaufnahme, and a five-vowel proto-system — that cannot be explained by chance or contact alone. We further propose that the Tyrsenian branch (proto-Etruscan/Pelasgian/Minoan) reached the Aegean during the Early Bronze Age Kura-Araxes expansion (c. 3100–2500 BC), eventually establishing itself in Italy as a remote colony that survived Bronze Age Indo-Europeanization to become the historical Etruscans. We additionally incorporate Kassite as a probable eastern branch of the CHG linguistic family, extending the CHG linguistic arc from the Zagros mountains in the east to the Italian peninsula in the west.

1. Introduction

The linguistic prehistory of the ancient Near East and Mediterranean remains one of the most contested areas of historical linguistics. Several language families and isolates — Hurro-Urartian, Kartvelian, Kassite, Minoan, and Etruscan — resist classification within any established macro-family and have long been treated as isolates or near-isolates. Standard comparative methodology, hampered by small corpora, poor attestation, and millennia of linguistic change, has struggled to identify genealogical relationships among these families.

In parallel, a revolution in ancient DNA analysis has transformed our understanding of prehistoric population movements. Crucially, the identification of Caucasian Hunter-Gatherer (CHG) ancestry as a distinct and geographically coherent genetic component, present in high concentrations among Kartvelian-speaking populations and progressively decreasing westward and southward, provides a new framework for interpreting linguistic relationships. If language tends to follow population continuity over deep time, then the CHG genetic world — extending from the South Caucasus through the Armenian Highland and eastern Anatolia — should correspond to a CHG linguistic world.

This paper develops and defends that hypothesis, proposing that Hurro-Urartian, Kartvelian, and Tyrsenian are all descendants of a common CHG Mesolithic ancestor, extended by the probable inclusion of Kassite as an eastern branch. We support this claim through four independent lines of evidence: personal pronoun comparison, grammatical structure, phonological inventory, and archaeogenetic data on population movements. We further integrate recent findings on the Kura-Araxes culture to propose a comprehensive model of how these language families reached their historical locations.

2. Background: The CHG Linguistic Hypothesis

2.1 Caucasian Hunter-Gatherers: Genetic Identity

CHG populations diverged from Western Hunter-Gatherers (WHG) probably during the Last Glacial Maximum, surviving in isolation in the South Caucasus refugium for millennia before being identified in the archaeogenetic record. Ancient DNA studies have shown that CHG ancestry is highest among modern Kartvelian-speaking populations, particularly western Georgians, who preserve the strongest genetic signal of this Mesolithic component. A landmark Bayesian phylogenetic study published in *Scientific Reports* (Gavashelishvili et al. 2023) confirms this deep continuity, placing the Kartvelian Urheimat in the Colchis glacial refugium of the western South Caucasus and demonstrating — through combined analysis of proto-Kartvelian wildlife vocabulary, archaeological biome distributions, and population genetics — that proto-Kartvelian was spoken in the South Caucasus by hunter-gatherers prior to the Neolithic, at least before 8000 BP and plausibly well before that. Critically, the study finds that CHG-related ancestry has been the dominant genetic component of South Caucasian populations continuously since at least Neolithic times, confirming that the Kartvelian linguistic

tradition has deep pre-Neolithic roots in the region rather than being a later Neolithic or Bronze Age arrival (Gavashelishvili et al. 2023). CHG ancestry is also detectable in Armenian Highland populations, Anatolian Chalcolithic groups, and — crucially — in Early Bronze Age Aegean populations, where a measurable influx of CHG-related ancestry is documented beginning around 3100–2500 BC (Lazaridis et al. 2022).

The geographic distribution of CHG ancestry thus traces a broad arc from the Caucasus westward through Anatolia into the Aegean, and southward into the Zagros mountains of Iran. This arc corresponds strikingly with the distribution of the language families discussed in this paper.

2.1a The ANE Component and Deep-Time Linguistic Connections

The CHG genetic profile was not formed in complete isolation. Crucially, CHG populations carry an approximately 10% Ancient North Eurasian (ANE) ancestry component — a gene flow arriving from the north, from populations related to Upper Palaeolithic Siberians of south Siberia or central Asia, most closely represented by the AfontovaGora3 individual from Lake Baikal (~17,000 BP). This ANE component is attested already in the oldest CHG individual: the Satsurblia cave specimen from western Georgia dated to approximately 13,300 BP, which represents the oldest fully formed CHG genetic profile and is genetically closest to modern South Caucasian (Kartvelian-speaking) populations (Jones et al. 2015). The Kotias Klde specimen dated to ~9,700 BP confirms continuity of this profile through the Mesolithic.

This ANE gene flow from the north into the CHG world has significant linguistic implications. ANE-related ancestry is also the major component of Eastern Hunter-Gatherers (EHG), who are the primary genetic substrate of proto-Uralic populations. The presence of ANE ancestry in the CHG world — documented already at 13,300 BP — means that CHG and EHG populations shared a common ancient genetic thread reaching back to Upper Palaeolithic Siberian populations. This shared ancestry provides a plausible mechanism for the deep-time linguistic similarities observed between the CHG language families (particularly the M-T pronoun pattern) and proto-Uralic and proto-Indo-European: these resemblances may reflect an ancient common ancestor or prolonged contact zone between ANE-derived populations in the north and CHG populations in the south Caucasus, predating both the Neolithic and the formation of the modern language families. The Caucasus mountains thus functioned not as a barrier but as a corridor connecting the CHG world with the northern Eurasian linguistic sphere in deep prehistory (Mattiangeli et al. 2019).

2.2 The Families Under Discussion

Hurro-Urartian (HU) is an extinct family comprising Hurrian (attested c. 2300–1200 BC across northern Mesopotamia and the Armenian Highland) and Urartian (attested c. 900–600 BC in the Van region). The family is morphologically agglutinative, strictly suffixing, and ergative-absolutive in alignment. It has long been treated as a language isolate, with proposed connections to Northeast Caucasian (the 'Alarodian' hypothesis) remaining controversial.

Kartvelian is the sole surviving Caucasian family with unambiguous CHG genetic association. It comprises Georgian, Mingrelian, Laz, Svan, and several other languages spoken in the Republic of Georgia and neighboring areas. Proto-Kartvelian is reconstructed with a five-vowel system, a three-way stop contrast (voiced/voiceless/ejective), and a rich case system. Verbs are polypersonal and both prefix- and suffix-based.

Tyrsenian is the proposed family comprising Etruscan (Italy, c. 700–100 BC), Lemnian (Aegean island of Lemnos, c. 600 BC), and Raetic (Alps, c. 500–100 BC). All three share distinctive case endings, verbal morphology, and a four-vowel system. Etruscan is the best-attested member with over 10,000 inscriptions. Lemnian is known from only two main inscriptions and scattered graffiti.

3. Personal Pronoun Comparison

3.1 The M-T Pronominal Zone

One of the most informative typological features in Eurasian linguistics is the distribution of the so-called M-T pronoun pattern, in which the first person singular pronoun is built on the consonant *m and the second person singular on *t. This pattern is shared by Proto-Indo-European (*me- / *tu-), Proto-Uralic (*mi / *ti), Proto-Kartvelian (*men / *sen), and Etruscan (*mi / *un). Its distribution across geographically dispersed families suggests either deep common ancestry or an ancient areal feature of Mesolithic Eurasia, possibly originating at the CHG-EHG (Eastern Hunter-Gatherer) contact zone north of the Caucasus.

Pers on	PIE	Proto -Urali c	Proto-Kar tvelian	Hurrian	Etruscan	Lemnian
1sg	*egh / *me-	*mi	*men	*iste/isti	*mi	*amoi-
2sg	*tu- / *te-	*ti	*sen	*fe/ve	*un	*-
3sg	(demonstrative)	*se/ta	*ima / *ege	*ane	*eca/ica	*-
1pl	*nos / *we-	*me	*cwen	*istena	*eni	*-
2pl	*yu- / *wos	*te	*tkwen	*fena	*-	*-

3.2 Kartvelian and Etruscan: The Exact Match

The most striking pronominal correspondence in the entire comparison is between Etruscan and Kartvelian. The Etruscan first person singular pronoun *mi is attested on hundreds of objects and corresponds exactly to Svan *mi and Georgian *me (reduced from *men). The Etruscan third person singular *eca corresponds to Svan *eja and

Georgian *ege/igi. This is not a superficial resemblance — both languages use these same roots in identical grammatical functions. Their presence in both families is most parsimoniously explained by common CHG ancestry, rather than by contact, given the geographic separation between Etruria and the Caucasus.

3.3 Hurrian as the Anomaly: A Post-Split Contact Event

Hurrian's pronoun system — 1sg *iste, 2sg *fe/ve, 3sg *ane — matches none of the other families in this study. The 1sg *iste has no parallel in any Caucasian or Eurasiatic family. The 2sg *fe/ve with a labiodental initial is typologically unusual globally. This pronominal isolation is best explained not by Hurrian being an unrelated language, but by a contact event in which an unattested language replaced the original CHG-derived M-T pronouns of proto-Hurrian, while leaving the deeper structural skeleton (suffixing morphology, ergativity, Suffixaufnahme) intact.

Critically, this pronominal replacement must have occurred after the split of proto-HU from proto-Tyrsenian — that is, after approximately 3500 BC — because Tyrsenian (Etruscan) preserves the M-T pronoun (*mi 'I') that Hurrian has lost. Had the contact event predated the HU-Tyrsenian split, Tyrsenian would also have lost its M-T pronouns. The fact that it did not means the unknown contact language encountered proto-Hurrian specifically in the Armenian Highland, after proto-Tyrsenian had already departed westward toward the Aegean. The most likely candidates for this contact language are either a now-extinct Caucasian family from the northeastern Caucasus, or an unknown population of the Armenian Highland that has left no surviving linguistic descendants. The parallel with Proto-Indo-European is instructive: PIE similarly preserved an Eurasiatic/Uralic pronominal substrate (the M-T pattern) while acquiring NW Caucasian phonological features through contact — a hybrid formation process that produced a language whose mixed ancestry resists clean genealogical classification. Proto-Hurrian underwent an analogous process in reverse: it retained CHG structural features while losing CHG pronouns through contact, producing a language that looks like an isolate to traditional classification.

An alternative explanation — requiring no external contact — merits serious consideration. Examining the Hurrian paradigm closely reveals a striking internal regularity: 1sg *isti* and 1pl *istena* share the same root, as do 2sg *fe* and 2pl *fena*. The plural forms are simply the singular stem plus the Hurrian animate plural suffix *-na*, which is the same suffix used productively across all animate nouns in the language. This pattern — absent in Kartvelian, where singular and plural pronoun roots are suppletive — suggests a possible internal mechanism: at some point in proto-HU history, the pronominal plural forms were rebuilt by productively attaching *-na* to some available stems (perhaps demonstratives or other deictic elements). Once those new plural forms (*istena, fena*) became established, analogical back-formation pressure — extremely well attested in agglutinative languages — generated new singular forms by stripping the familiar *-na* suffix: *istena* → *isti*, *fena* → *fe*. The original M-T singulars (*men, sen*) were then displaced by these internally derived forms. Both steps — productive extension of a nominal plural suffix into the pronominal paradigm, and back-formation of singular stems from plurals — are individually well-attested mechanisms of language-internal change, and their combination requires no borrowing

from an unknown source language. This hypothesis is further supported by the fact that the Hurrian plural pronouns preserve a closer phonological relationship to the Kartvelian plurals (*cwen, tkwen*) than the singulars do, consistent with the plurals being the more conservative forms. The two hypotheses — external contact replacement and internal back-formation — are not mutually exclusive; a period of language contact could have facilitated the paradigm restructuring even if the specific forms were internally generated.

4. Grammatical Structure Comparison

4.1 Core Typological Features

Feature	Hurro-Urartian	Kartvelian	Tyrsenian	Kassite (inferred)
Morphological type	Agglutinative	Agglutinative	Agglutinative	Agglutinative (inferred)
Primary morphological direction	Pure suffixing	Both (prefix-heavy verbal)	Pure suffixing	Suffixing (inferred)
Alignment	Ergative-absolutive	Ergative (split)	Active-stative/ergative	Unknown
Grammatical gender	No — animacy distinction	No	No — animacy distinction	Unknown
Animacy-based nominal classes	Yes (animate/inanimate)	Partially	Yes (animate/inanimate)	Probable
Suffixaufnahme	Yes — hallmark feature	Yes	Partial evidence	Unknown
Case system richness	8–15 cases	7+ cases	6 cases (Etruscan)	Unknown
SOV word order	Strict	Strict	SOV	Unknown
Verb agrees with object	Yes	Yes (polypersonal)	Partially	Probable
Postpositions vs prepositions	Postpositions	Postpositions	Postpositions	Unknown

4.2 Suffixaufnahme: The Diagnostic Isogloss

Suffixaufnahme — the phenomenon by which dependent modifiers of a noun copy the noun's own case suffixes — is typologically very rare globally and is among the most powerful genealogical diagnostics available for these language families. Its presence in Hurro-Urartian and Kartvelian is well-documented. Its partial presence in Etruscan has been noted by several scholars. The geographic distribution of Suffixaufnahme — covering precisely the arc of CHG-adjacent populations from the Armenian Highland to the Caucasus to the Aegean — is unlikely to be coincidental and strongly supports a common CHG linguistic origin for all families sharing this feature.

4.3 The Pure-Suffixing Architecture: HU and Tyrsenian vs. Kartvelian

A critical distinction within the CHG cluster separates Kartvelian from both HU and Tyrsenian. Kartvelian develops a massively prefix-heavy verbal morphology — Georgian verb forms can incorporate up to six morpheme slots before the root. HU and Etruscan/Lemnian are by contrast almost purely suffixing, with no significant verbal prefixing. This architectural difference likely reflects a post-split innovation in Kartvelian driven by its specific morphological history in the Caucasus. HU and Tyrsenian, having separated from the CHG ancestor and moved southward and westward respectively, preserved the original suffixing template without developing the prefix-heavy verbal system.

This shared suffixing architecture between HU and Tyrsenian, absent in Kartvelian, constitutes a secondary isogloss linking these two westward-moving branches of the CHG family against the northern Caucasian branch.

5. Phonological Comparison

5.1 The Proto-CHG Phonological System

Reconstructing the proto-CHG phonological system requires comparison across all three daughter families, identifying shared retentions against family-specific innovations. The following features appear in all three families and are proposed as CHG retentions:

A five-vowel system /a, e, i, o, u/ at the proto-level — preserved completely in Kartvelian, partially in HU (with dialectal merger of o/u in Hattusha Hurrian), and reduced to four vowels in attested Tyrsenian (Etruscan lost /o/, Lemnian appears to retain /o/ but loses /u/). The east-to-west vowel reduction gradient is consistent with progressive phonological change away from the CHG homeland.

SOV-driven phonotactics with constraints on initial consonant complexity — both HU and Tyrsenian restrict initial consonant clusters, while Kartvelian has developed massive initial clusters as a secondary innovation driven by its prefix-heavy verbal morphology.

Phonemically distinctive consonant length (geminales) — present in HU and partially in Tyrsenian, absent in Kartvelian, suggesting this feature was present in the CHG ancestor and lost in the northern (Kartvelian) branch.

5.2 Phonological Comparison Table

Feature	Proto-Kartvelian	Hurro-Urartian	Tyrsenian (Etruscan/Lemnian)
Vowel inventory	5: a, e, i, o, u + length	5: a, e, i, o, u (Mitanni); 3-way merger in Hattusha	4: Etruscan a/e/i/u; Lemnian a/e/i/o
Vowel length phonemic	Yes	Yes	Not phonemic in Etruscan
Stop series	Voiced / Voiceless / Ejective	Allophonic voicing only; emphatics probable ejectives	Voiceless / Aspirated (no voiced stops)
Ejectives	Yes — hallmark CHG feature	Probably in proto-HU; lost in Hurrian	No — replaced by aspiration
Initial consonant clusters	Yes — massively developed	No — r/l prohibited initially	No — few initial clusters
Geminate consonants	No phonemic geminates	Yes — phonemically distinctive	Partial — in morphological contexts
Pharyngeals/laryngeals	Yes in proto-language	h-sounds attested	h attested; no pharyngeals
Stress	Not word-initial dominant	Unknown	Word-initial dominant (drives syncope)

5.3 The Vowel Reduction Gradient

The most diagnostically informative phonological comparison concerns the five-vowel proto-system. Kartvelian preserves all five vowels throughout its history, with Svan even adding additional vowels. Hurrian preserves five vowels in the older Mitanni dialect but

shows o/u merger in the later Hattusha dialect — an internal dialectal innovation showing exactly the same merger direction that Tyrsenian carries to completion. Etruscan completes the loss of /o/ entirely, retaining only four vowels. Lemnian, more archaic than Etruscan, preserves /o/ but loses /u/ in its attested forms.

This gradient — full five vowels (Kartvelian) > five vowels with incipient dialectal merger (Mitanni Hurrian) > four vowels through o-loss (Etruscan) — runs precisely along the east-to-west geographic axis of the proposed CHG dispersal, from the Caucasian homeland toward the western terminus in Italy. This is not expected under a contact or coincidence model, but is a natural consequence of progressive phonological drift with increasing geographic distance from the origin point.

5.4 The Stop System Divergence

The most dramatic phonological difference between the three families concerns the treatment of the stop system. Proto-Kartvelian preserves the canonical three-way Caucasian contrast: voiced / voiceless / ejective. This is widely recognized as a CHG inheritance, as ejective-heavy phonologies are characteristic of Caucasian languages broadly. Hurrian loses the three-way distinction in its attested form — voicing becomes allophonic, though Urartian preserves what may be ejectives as 'emphatics'. Tyrsenian goes furthest from the CHG norm: voiced stops are absent, ejectives are replaced by an aspirated series (ph, th, kh), producing a system phonologically convergent with Greek through millennia of Aegean contact.

The directionality is again west-of-origin = most innovated. Kartvelian, staying in the Caucasus, preserves the CHG ejective system. HU, moving south, partially retains it. Tyrsenian, moving furthest west, replaces it entirely with aspiration under Aegean contact influence.

6. The CHG Linguistic Arc: Geographic and Temporal Reconstruction

6.1 The Full CHG Language Family

Based on the structural, pronominal, and phonological evidence assembled in this paper, we propose that the following language families constitute branches of the CHG macro-family, arranged geographically from north to south/west:

Branch	Geographic Zone	Attested Period	Status	CHG Evidence
Kartvelian	South Caucasus	Medieval to present	Surviving	Highest CHG ancestry; Urheimat in Colchis glacial refugium confirmed pre-Neolithic

Branch	Geographic Zone	Attested Period	Status	CHG Evidence
				(Gavashelishvili et al. 2023); M-T pronouns; ejective phonology
Hurro-Urartian	Armenian Highland, N. Mesopotamia	c. 2300–600 BC	Extinct	CHG-adjacent homeland; Suffixaufnahme; suffixing morphology; animacy classes
Kassite	Zagros mountains, Babylonia	c. 1600–1155 BC	Extinct (48 words only)	Geographic position at CHG eastern boundary; divine name parallels with HU; onomastic parallels with Kartvelian
Tyrsenian	Aegean, Italy, Alps	c. 600 BC – 100 AD	Extinct	Exact M-T pronoun match with Kartvelian; Suffixaufnahme; suffixing morphology; archaism gradient Lemnian > Etruscan toward HU

6.2 The Split Chronology

The temporal sequence of splits within the CHG macro-family can be estimated by comparing linguistic distance with known migration events and archaeogenetic data:

The Kartvelian-HU split is estimated to have occurred at least as early as approximately 8000 BC. This split was not merely a neutral demographic divergence — it was almost certainly forced and made permanent by the arrival of early Neolithic farmers into the South Caucasus region, most probably speakers of a proto-Northeast Caucasian language tradition expanding from the eastern Anatolian and Zagros Neolithic frontier. This farming expansion, beginning around 8500–8000 BP, penetrated the South Caucasus from the south, introducing agriculture into what had been CHG hunter-gatherer territory and effectively creating a linguistic barrier between two surviving CHG-speaking refugia. The western refugium, sheltered on the southern slopes of the Caucasus mountains and in the densely forested Colchis region around the Rioni and Enguri watersheds, preserved proto-Kartvelian — exactly as confirmed by the Gavashelishvili et al. (2023) study. The eastern and southern refugium, pushed into the highland zone around lakes Van and Urmia in the Armenian Highland, preserved proto-HU-Tyrsenian. The Neolithic farmer expansion did not necessarily replace these

populations genetically but severed the geographic continuity between them, preventing further contact and allowing linguistic divergence to consolidate irreversibly. It is possible that the proto-CHG linguistic split had already begun earlier — the two branches may already have been dialects rather than a single language — but the Neolithic farmer barrier definitively ended any possibility of continued contact or convergence between them. The lakes Van and Urmia region subsequently became the heartland from which proto-HU expanded as the Kura-Araxes culture, while proto-Tyrsenian eventually split off and moved westward into the Aegean.

The HU-Tyrsenian split (if from a common intermediate ancestor) or the proto-Tyrsenian departure from the CHG world is estimated at approximately 3500–3000 BC, consistent with the beginning of the Kura-Araxes cultural expansion and the archaeogenetically documented influx of CHG-related ancestry into the Aegean beginning around 3100 BC. The linguistic distance between Hurrian and Etruscan — total pronominal divergence, extensive lexical divergence, but shared structural architecture — is consistent with approximately 4000 years of separation before attestation.

The Etruscan-Lemnian split is estimated at approximately 2500–2000 BC — a significantly shallower split than HU-Tyrsenian, based on the near-identical case morphology and verbal endings shared by both Tyrsenian languages, with the differences (Lemnian preserves archaic *-ai verbal endings and /o/ vowel lost in Etruscan) being consistent with 1500–2000 years of separation before attestation.

The Etruscan-Raetic split is the shallowest, estimated at approximately 1500–1000 BC, consistent with the Villanovan/proto-Etruscan expansion northward into the Alps during the Italian Bronze Age. Shared vocabulary (*zal 'two', *clan 'son') and virtually identical case suffixes support a relatively recent common ancestor.

7. The Tyrsenian Migration: Kura-Araxes to the Aegean and Italy

7.1 The Kura-Araxes Cultural Expansion

The Kura-Araxes (Early Transcaucasian) culture, originating in the South Caucasus and Armenian Highland around 3500–3300 BC, underwent one of the most extensive expansions of the Chalcolithic-Early Bronze Age period, reaching eastern Anatolia, the Levant, the Zagros, and the Aegean. Archaeogenetically, Kura-Araxes populations carry approximately 70% CHG ancestry, making them the primary vector by which CHG genetic material entered the broader Near Eastern and Aegean world during the Early Bronze Age.

Lazaridis et al. (2022) documented a measurable influx of CHG-related ancestry into Aegean populations beginning around 3100–2500 BC — precisely the period of the Kura-Araxes expansion and the emergence of Minoan civilization on Crete (Minoan EBA I beginning c. 3100 BC). We propose that proto-Tyrsenian was the language carried by this CHG-enriched population into the Aegean, and that Minoan civilization represents the Aegean florescence of this proto-Tyrsenian/Pelasgian cultural tradition.

7.2 The Pelasgian Identification

Ancient Greek sources consistently identify the pre-Greek population of the Aegean as 'Pelasgian', with Lemnos, Imbros, and Skyros cited as their last strongholds. Multiple independent ancient sources — Hellanicus of Lesbos, Strabo citing Anticlides, and Euripides — place the Pelasgians as the common ancestor of both the Lemnians and the Etruscans, with the directionality consistently given as Aegean-to-Italy rather than the reverse. While mythological narratives cannot be taken as literal history, their consistent directionality is informative. Linguistically, the directionality is confirmed by the archaism profile: Lemnian is measurably more archaic than Etruscan (preserving the *-ai verbal ending and /o/ vowel lost in Etruscan), consistent with the Aegean branch being the source and Italy being the destination, not the reverse.

7.3 The Italian Colony

The proto-Etruscan presence in Italy is proposed to derive from a Minoan/Pelasgian maritime colony established in central Italy, possibly during the Italian Chalcolithic or Early Bronze Age (c. 2500–2000 BC). This is broadly consistent with the appearance of Aegean-connected material culture in central Italy during this period. The colony would have overlaid the existing Cardial Ware population without significant genetic replacement, as the genetics of the Rinaldone culture (pre-Bell Beaker Italy) show essentially pure Early European Farmer (EEF) ancestry with no detectable Kura-Araxes component.

This apparent contradiction — linguistic replacement without genetic replacement — is resolved by recognizing that the proto-Etruscan arrival may have been demographically small but culturally and linguistically prestigious, with language shift occurring without population replacement. This mechanism is well-documented historically: the spread of Latin, Aramaic, and Arabic provide clear examples of linguistically dominant minorities shifting the language of demographically larger populations.

7.4 Bell Beaker Survival

The subsequent arrival of Bell Beaker and Bronze Age steppe populations in Italy, bringing Indo-European languages, genetically transformed the Italian population while the proto-Etruscan language survived in Etruria specifically — possibly because Etruscan cultural prestige, urban complexity, or geographic concentration in the Tyrrhenian zone made linguistic replacement less effective there than in surrounding regions. However, this does not mean proto-Etruscan was immune to IE influence. Prolonged coexistence with early Indo-European-speaking populations during the Italian Bronze Age (c. 2200–900 BC) left measurable traces in Etruscan vocabulary, particularly in terms relating to metallurgy, social organization, and trade — domains where the technologically innovative IE-speaking Bell Beaker and later Terramare/Villanovan-adjacent cultures were dominant. Some Etruscan divine names and institutional terminology show possible IE substrate influence, reflecting the reality that proto-Etruscan absorbed IE loanwords across centuries of bilingual contact even while preserving its non-IE grammatical structure. This IE vocabulary layer in Etruscan has sometimes confused scholars into proposing IE affiliation for Etruscan, but is better

understood as a contact stratum overlying the genuine CHG-descended linguistic core. The Raetic differentiation follows as the proto-Etruscan community expanded northward into the Alps, producing a linguistically divergent Alpine branch that was eventually absorbed by Latin and Celtic expansion.

8. The Two-Layer Pre-IE European Substratum

8.1 Cardial Ware as an NE Caucasian-Adjacent Tradition

A critical clarification emerges from the revised chronology of European Neolithic expansion. The Cardial Ware / Impressed Ware culture, which brought agriculture to the western Mediterranean beginning around 6400 BC (earliest sites in Epirus and Corfu), is genetically dominated by Early European Farmer (EEF) ancestry with only trace CHG. This population derives primarily from Anatolian Hunter-Gatherer (AHG) ancestry through the western Anatolian Neolithic transition. It is therefore linguistically distinct from the CHG-adjacent populations that produced Hurro-Urartian, Kartvelian, and Tyrsenian.

The language of the Cardial Ware farmers is proposed here to have been ancestral to Basque — the sole surviving language of this Neolithic agricultural stratum in western Europe. This assignment is supported by the well-documented structural and lexical connections between Basque and Northeast Caucasian (NE Caucasian / Nakh-Dagestanian), which are most naturally explained as a common ancestor shared between the Cardial farming tradition and the NE Caucasian-adjacent populations of the eastern Anatolian/Zagros Neolithic frontier from which the Cardial farmers ultimately derived their agricultural package.

8.2 The Two-Layer Model

Layer	Culture	Date	Genetic Signature	Proposed Language Family	Survivors
Layer 1	Cardial / Impressed Ware Neolithic	c. 6400–4000 BC	EEF dominant (AHG-derived)	NE Caucasian-adjacent (proto-Basque)	Basque (sole European survivor)
Layer 2	Kura-Araxes / Pelasgian EBA	c. 3100–2500 BC	CHG-enriched (70% CHG)	Proto-Tyrsenian / Pelasgian	Etruscan (Italy), Lemnian (last Aegean), Minoan/Eteocretan (Crete)

Layer	Culture	Date	Genetic Signature	Proposed Language Family	Survivors
Layer 3	Bell Beaker / Yamnaya	c. 2500–1800 BC	Steppe-dominant (EHG+CHG)	Proto-Indo-European branches	All modern IE languages of Europe

This two-layer model for the pre-IE European linguistic substrate is more parsimonious than existing models that either treat all pre-IE languages as a single family or treat each as an independent isolate. It distributes the substrate languages into two geographically and chronologically distinct migration events, each with a clear genetic correlate, and makes the survival of Basque and Etruscan as the two pre-IE isolates of historical Europe a natural consequence of geographic isolation (Basque in the Pyrenean refuge; Etruscan in the Tyrrhenian zone) rather than an improbable coincidence.

9. Kassite: The Eastern Extension of the CHG Arc

9.1 Kassite as an Eastern Branch

Kassite, the language of the Kassite dynasty that ruled Babylonia from approximately 1600 to 1155 BC, is attested in only 48 glosses in a bilingual Kassite-Babylonian vocabulary plus royal names. Despite this minimal corpus, Fournet (2011) identified structural and lexical parallels with HU that are difficult to dismiss as coincidental: the storm god name (Kassite *Tispak ~ Hurrian *Tessub ~ Uartian *Teissaba); core vocabulary (*yas 'land' ~ Hurrian *ese 'earth'); and shared personal name formation patterns (Verb + divine Name). The geographic origin of the Kassites in the Zagros mountains places them at the eastern boundary of CHG population territory. We propose Kassite as the eastern branch of the CHG macro-family, extending the CHG linguistic arc from the Zagros to the Italian peninsula.

10. The Lemnian Archaism Test: Directionality of Tyrsenian Dispersal

10.1 Lemnian as More Archaic Than Etruscan

A key prediction of the east-to-west dispersal model is that the Lemnian branch (closer to the Aegean source area) should preserve more archaic features than the Italian Etruscan branch (the western terminus of the dispersal). This prediction is confirmed by the available evidence on multiple independent features:

First, the verbal perfect ending: Lemnian preserves *-ai as a productive verbal perfect/preterite ending (e.g. *aomai 'I am'), while Etruscan has replaced this with

*-ce/-ke. The *-ai ending is independently attested in Hurrian as a modal/jussive morpheme, suggesting it is the older form. Etruscan's *-ce is the innovation. Crucially, Lemnian attests both *-ai and *-ke simultaneously, caught at an intermediate stage of the *-ai > *-ce transition that Etruscan has completed.

Second, the vowel system: Lemnian preserves the vowel /o/ that Etruscan has completely lost. Since Hurrian (Mitanni dialect) also retains /o/, the Lemnian /o/ is the archaic form and Etruscan /u/ (replacing /o/) is the innovation.

Third, the genitive morpheme: Lemnian shows more consistent use of the genitive *-s, while Etruscan has developed allomorphic variation (*-s/-as/-es/-is/-us/-ls), suggesting greater phonological erosion in the western Italian branch.

The combination of multiple independent archaisms in Lemnian pointing toward HU — the proposed eastern source — and multiple independent innovations in Etruscan pointing away from HU — the proposed western terminus — constitutes strong evidence for the east-to-west dispersal model and effectively rules out the mainstream alternative of a westward Etruscan colonial foundation of Lemnian.

10.2 Refuting the Colonial Model

The mainstream position — that Etruscan traders or pirates established a language colony on Lemnos in the 7th–8th century BC, transmitting their language to the island — is refuted by the archaism profile. A colony established in the 8th century BC would speak contemporary Etruscan, not an older form of it. The fact that Lemnian features consistently correspond to the earliest attested phases of Etruscan — noted by specialists themselves — demands a split predating classical Etruscan, which itself predates the proposed colonial foundation date. The colonial model is also refuted by the absence of archaeological evidence for an Etruscan settlement on Lemnos, despite extensive excavation at the site of Hephaistia.

The adoption of a southern Etrurian variant of the Etruscan alphabet for writing Lemnian is best explained not as evidence of colonial foundation but as an Iron Age script-contact event between long-separated linguistic relatives: Etruscan maritime activity in the Aegean brought the Etruscan script into contact with a population whose linguistic kinship with Etruscans was recognized, prompting adoption of the Etruscan-variant alphabet in preference to the surrounding Greek scripts.

11. Discussion: Implications and Open Questions

11.1 The M-T Pronoun as a CHG-EHG Contact Feature

The M-T pronoun pattern shared by Kartvelian, Etruscan, PIE, and Uralic raises a question about its ultimate origin. If it is a CHG inheritance, why do PIE and Uralic — families with minimal or no CHG genetic ancestry in their core — also share it? We propose that the M-T pattern is not specifically CHG but rather a feature of the CHG-EHG contact zone north of the Caucasus during the Mesolithic, where proto-Uralic/Eurasiatic speakers interacted with CHG populations. The M-T pronoun

was shared across this contact frontier, which is why it appears in both CHG descendants (Kartvelian, Etruscan) and EHG descendants (Uralic, PIE). Hurrian's departure from M-T represents a specific contact event replacing the inherited pronouns — an event without parallel in the other CHG branches.

11.2 The Role of Minoan

Minoan, the undeciphered language of Bronze Age Crete, is predicted by this framework to be a branch of the Tyrsenian family — the Aegean island dialect of the same proto-Tyrsenian tradition that reached Italy as Etruscan and survived on Lemnos as Lemnian. Eteocretan, the post-Minoan language of Crete attested in a few Iron Age inscriptions, would then be the direct descendant of Minoan. This prediction is partially testable: Linear A's statistical properties (morpheme length distribution, suffix-heaviness, word boundary patterns) should be consistent with an agglutinative suffixing language if Minoan is Tyrsenian. Computational analyses of Linear A's structural properties are consistent with this expectation, though phonetic decipherment remains elusive.

11.3 The Alarodian Hypothesis Reconsidered

The Alarodian hypothesis — proposing a genetic relationship between Hurro-Urartian and Northeast Caucasian — is not incompatible with the CHG framework proposed here, but requires reinterpretation. Rather than HU being a branch of NE Caucasian, or NE Caucasian of HU, the structural similarities between HU and Nakh-Dagestanian languages are more parsimoniously explained as contact borrowing — specifically, extensive lexical and structural borrowing from Hurrian into the Nakh branch of NE Caucasian during the Kura-Araxes period, when Hurrian-adjacent and proto-Nakh populations shared the eastern Anatolian/Armenian Highland landscape for millennia. This contact explanation is supported by the uneven distribution of Alarodian parallels: they cluster in Nakh specifically, rather than being evenly distributed across NE Caucasian as expected from true common ancestry.

11.4 Outstanding Problems

Several problems remain unresolved and are acknowledged as limitations of the current proposal. The mechanism of proto-Etruscan arrival in Italy without detectable genetic signal requires more detailed modeling of language shift scenarios. The complete replacement of the Hurrian pronominal system, while paralleled by the PIE case, lacks direct evidence of the contact language responsible. The Kassite corpus of 48 words is too small for rigorous phonological reconstruction. Minoan remains undeciphered.

These limitations notwithstanding, the convergent evidence from pronoun typology, grammatical architecture, phonological gradients, and archaeogenetic population modeling creates a framework that is substantially more parsimonious than treating HU, Kartvelian, and Tyrsenian as three independent isolates with no historical relationship.

12. Conclusion

This paper has argued for the existence of a CHG macro-family comprising at minimum Kartvelian, Hurro-Urartian, and Tyrsenian, with Kassite as a probable eastern branch. The evidence for this grouping rests on four independent lines of argument: shared structural features (pure suffixing morphology, ergative alignment, animacy-based nominal classes, Suffixaufnahme) distributed precisely along the arc of CHG genetic ancestry; a five-vowel proto-phonological system that degrades progressively westward from Kartvelian through HU to Tyrsenian; a pronominal convergence between Kartvelian and Etruscan (exact *mi match) consistent with common ancestry; and the archaism profile of Lemnian relative to Etruscan, which confirms east-to-west directionality of the Tyrsenian dispersal.

We have further proposed a two-layer model for pre-IE European linguistic substrate: a Cardial Ware / NE Caucasian-adjacent layer (ancestral to Basque) dating to the initial Neolithic agricultural expansion c. 6400 BC, and a Kura-Araxes / proto-Tyrsenian layer dating to the Early Bronze Age CHG expansion c. 3100–2500 BC. This model distributes the pre-IE isolates of historical Europe into two genetically and chronologically distinct events, resolving the longstanding puzzle of why Basque and Etruscan exist as apparently unrelated isolates in the same pre-IE substrate zone.

The CHG linguistic world, extending from the Caucasian mountains in the north to the Zagros in the east and the Italian peninsula in the west, represents one of the most geographically extensive and temporally deep linguistic traditions recoverable from the pre-literate world. Its partial recovery, through the convergent analysis of historical linguistics, typology, and archaeogenetics, illustrates the potential of multidisciplinary approaches to resolve questions that neither field could address alone.

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